

Christopher Farber

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EDUCATION

Purdue University <i>Master of Science in Computer Science</i> • Selected Coursework: Statistical Machine Learning, Natural Language Processing, Computer Vision with Deep Learning	Fall 2026 (Expected) West Lafayette, IN
Purdue University <i>Bachelor of Science in Computer Science (3.80 GPA)</i> • Selected Coursework: Systems Programming, Analysis of Algorithms, Data Mining & ML, Intro to AI	Fall 2025 (Expected) West Lafayette, IN

PROFESSIONAL EXPERIENCE

Purdue University <i>Undergraduate Research Assistant</i> • Applying ML methods to predict mortality outcomes in DMD patients using CMR data, in collaboration with the NIH. • Developing and evaluating UQ methods for multi-fidelity neural operator models, collaborating with Sandia National Labs. • Responsible for model development and experimental analysis under supervision of Dr. Guang Lin, Purdue University.	September 2025 - Present West Lafayette, IN
Kohl's <i>Software Engineering Intern</i> • Engineered a full-stack React/TypeScript + Kotlin/Spring Boot application to streamline the creation, sharing, and execution of test suites, significantly reducing test setup time for Kohl's Technology teams. • Upgraded 4 production-level repositories to use the new version of Kohl's' GitLab DevOps pipeline. • Delivered a workshop on Framer Motion to an audience of 40+ people, showing how it can enhance the Kohl's landing page.	June 2025 - August 2025 Menomonee Falls, WI
iRODS <i>Software Engineering Intern</i> • Refactored React app for the Zone Management Tool to communicate with newly-developed HTTP API. • Developed a TypeScript wrapper library for the HTTP API, consisting of 120+ types and 60+ unit tests written with Jest.	June 2024 - August 2024 Chapel Hill, NC
National Transportation Consultants <i>Software Engineering Intern</i> • Built an Azure AI model to detect and extract fields from 10+ document types, achieving 2x processing speed. • Coordinated with CEO and software development team to integrate model into company's operations.	June 2023 - August 2023 Fishers, IN
Primary Record <i>Software Engineering Intern</i> • Collaborated with company founders to create React + GCP web app to extract and categorize text on prescription labels. • Deployed as a feature in company's main app, allowing users to store prescription info with OCR.	June 2022 - August 2022 Fishers, IN

PROJECTS

Geospatial ML Group <i>Python, PyTorch, GIS</i> • Trained a ConvLSTM model with PyTorch to predict levels of PM2.5 and correct the bias of NASA's GEOS-FP model. • Collaborating with 2 NASA scientists to integrate model into the MAIA satellite project. • Established a data to training pipeline, enabling group to get a headstart on model analysis. • Presented poster at Purdue's 2024 GIS Day conference.	August 2024 - Present
Project oCEANIC <i>Python, Flask, Redis, AWS S3, Next.js, Leaflet.js</i> • Developed agent-based seafaring model using archaeological and paleoclimate data, revealing patterns of historical seafaring. • Preprocessed and saved 40 years of hourly NetCDF wind data from NASA's MERRA-2 dataset into an AWS S3 bucket. • Created and deployed Next.js website that allows users to visualize, interact with, and export simulations from the model. • Deployed Flask server that runs simulations for the website, along with a Redis server to store simulation data. • Presented preliminary research results at a workshop at the University of Hawaii Hilo, research talks at Purdue's 2024 & 2025 Fall Undergraduate Research Conferences, and posters at Purdue's 2024 & 2025 GIS Day Conferences.	January 2024 - Present
Analysis of Online HPV Vaccine Hesitancy <i>Python, Reddit API, pandas, NLTK</i> • Created a script to automatically collect data via Reddit's API from HPV-related subreddits each day, totaling in 316 posts. • Utilized scikit-learn to vectorize the data with TF-IDF, then K-Means and TSNE/SVD to find 3 significant clusters. • Applied NLP and statistical methods to analyze differences in sentiment and subjectivity across clusters, via NLTK. • Presented poster at Purdue's 2024 Spring Undergraduate Research Conference.	September 2023 - May 2024

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, Kotlin, R
Frameworks: Node.js, Next.js, Tailwind CSS, Jest, Cypress, Flask, Material-UI, PyTorch, Spring Boot
Developer Tools: Git, Google Cloud Platform, Microsoft Azure, AWS S3, Docker, nginx, CI/CD, Redis, Postgres
Libraries: React.js, Axios, pandas, NumPy, Matplotlib, TensorFlow, Keras, scikit-learn, NLTK, SpaCy